Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	LTVIP2026TMIDS78831
Project Name	Explore with AI: Custom Itineraries for Your Next Journey
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Input Handling	1. Enter destination through text input field 2. Enter number of days 3. Enter number of nights 4. Click “Generate Itinerary” button
FR-2	Input Validation	5. Validate all fields are filled 6. Validate days and nights are numeric 7. Restrict maximum trip length to 15 days 8. Ensure nights are less than number of days 9. Display warning messages for invalid inputs
FR-3	AI-Based Itinerary Generation	10. Build structured AI prompt dynamically 11. Send request to Gemini API 12. Generate day-wise itinerary 13. Generate travel tips section 14. Generate food recommendations 15. Generate hotel/stay suggestions
FR-4	Itinerary Display	16. Display generated itinerary in structured format 17. Show sections (Morning, Afternoon, Evening, Night) 18. Display headings clearly 19. Maintain emoji-based structured layout

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The system shall provide a simple and intuitive interface that allows users to generate itineraries without technical knowledge.
NFR-2	**Security**	The system shall securely manage API keys using environment variables or Streamlit secrets and not expose them in the frontend.
NFR-3	**Reliability**	The system shall handle API failures or invalid inputs gracefully without crashing.
NFR-4	**Performance**	The system shall generate the itinerary within a reasonable time (typically under 10–15 seconds depending on API response).
NFR-5	**Availability**	The system shall be accessible online via Streamlit deployment and available to users whenever the hosting service is running.
NFR-6	**Scalability**	The system shall support multiple users generating itineraries concurrently without affecting performance significantly.